



VMWARE VSPHERE 7.0: INSTALL, CONFIGURE, MANAGE



Lab 07: Creating and Managing vCenter Server Inventory

Document Version: 2020-11-04

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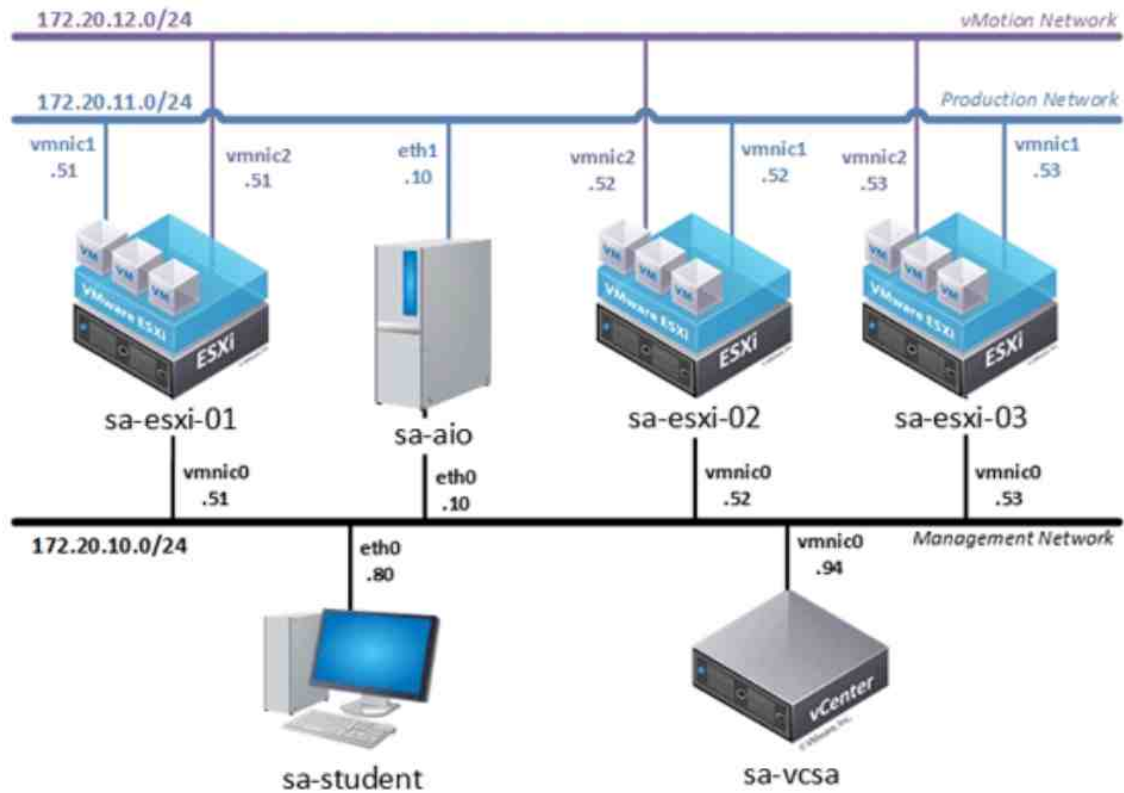
Introduction

In this lab, you will create and manage the vCenter Server inventory.

Objectives

-) Create Datacenter Object
-) Add your ESXi host to the vCenter Server Inventory
-) View information about the ESXi host
-) Configure the ESXi host as an NTP Client
-) Create a Folder for the ESXi host
-) Create Folders for Virtual Machines and Virtual Machine Templates

Lab Topology



Lab Settings

The information in the table below will be needed in order to complete the lab. The task sections below provide details on the use of this information.

Virtual Machine	IP Address	Account	Password
sa-esxi-01	eth0: 172.20.10.51	root	Train1ng\$
sa-esxi-02	eth0: 172.20.10.52	root	Train1ng\$
sa-esxi-03	eth0: 172.20.10.53	root	Train1ng\$
sa-aio	eth0: 172.20.10.10	sysadmin	vmware123
sa-student	eth0: 172.20.10.80	sysadmin	vmware123
sa-vcsa	eth0: 172.20.10.94	sysadmin@vclass.local	vmware123

1 Create a Datacenter Object

In this task, you will create a data center object named LAB-Datacenter to organize hosts and virtual machines within the VMware vSphere environment.

1. Launch the sa-student virtual machine to access the graphical login screen.

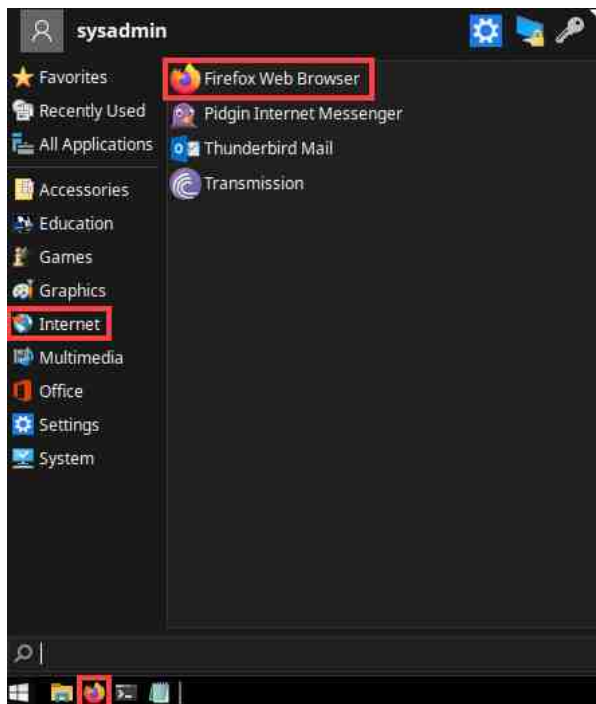


To launch the console window for a virtual machine, you may access by either clicking on the machine's graphic image from the topology page or by clicking on the machine's respective tab from the navigation bar.

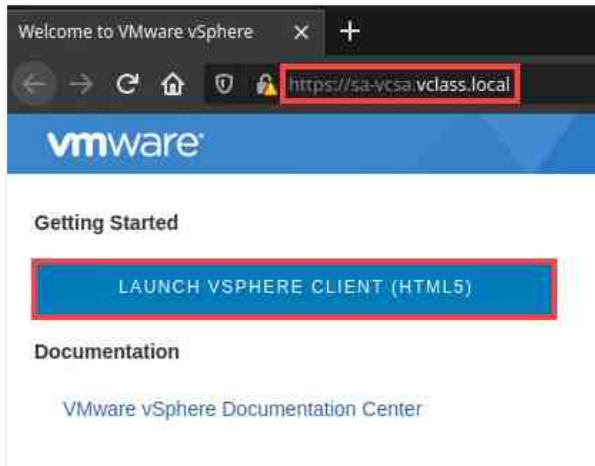
2. Log in as `sysadmin` using the password `vmware123`.



3. Launch the Mozilla Firefox web browser by either clicking on the icon shortcut found on the bottom toolbar or by navigating to Start Menu > Internet > Firefox Web Browser.

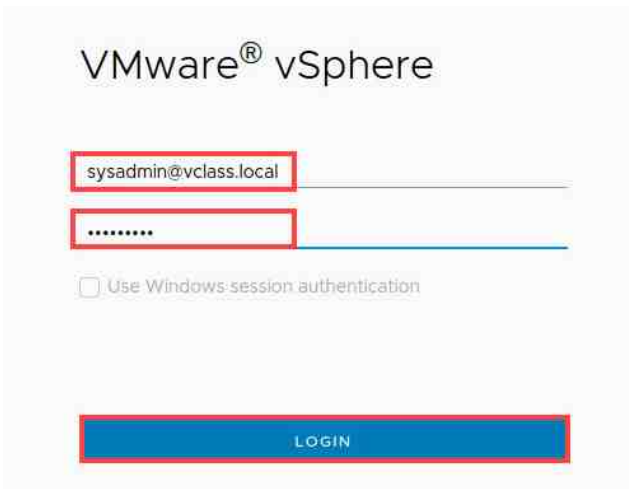


4. Notice the homepage is automatically configured to load the URL address of the VMware vCenter Server Appliance (sa-vcasa virtual machine). Click on the LAUNCH VSPHERE CLIENT (HTML5) button to launch the vSphere Client.



If you cannot successfully load the VMware Getting Started webpage, then please wait an additional 3-5 minutes and refresh the page to continue. During this time, the vCenter Server Appliance is still booting up and requires extra time to initialize.

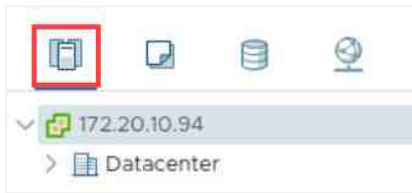
5. In the Username text field, type `sysadmin@vclass.local` and in the Password text field, type `vmware123`. Click on LOGIN.



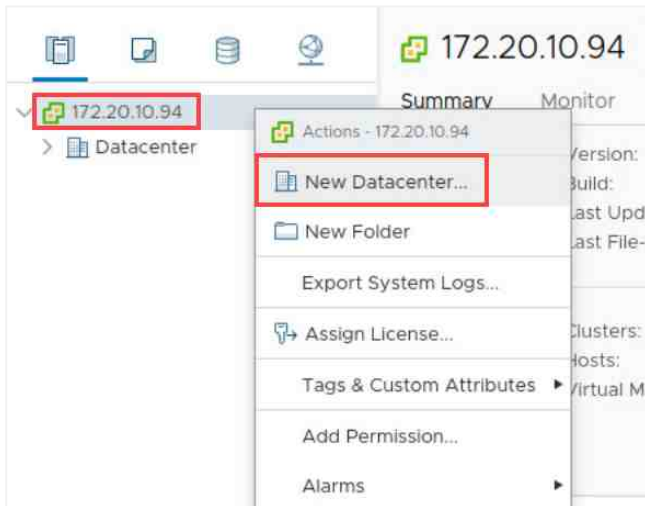
**Please
Note**

You may ignore the “browser-OS combination” warning message presented on the VMware vCenter Single Sign-On page and continue moving forward with the lab.

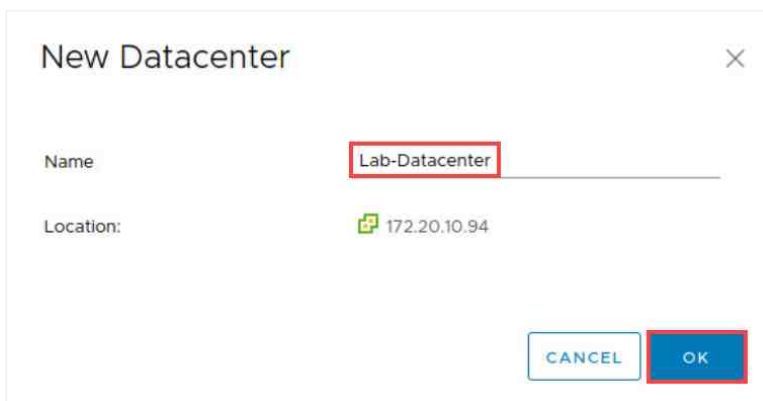
6. In the navigator pane, click on the Hosts and Clusters tab.



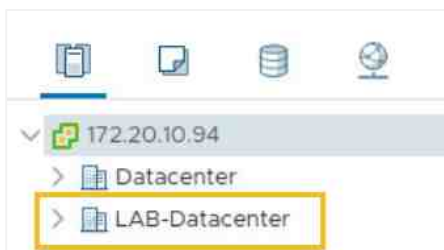
7. Right-click on the vCenter Server Appliance 172.20.10.94 and select New Datacenter.



8. In the New Datacenter window, type LAB-Datacenter for the Name and click OK.



9. In the navigator pane, verify that LAB-Datacenter appears.

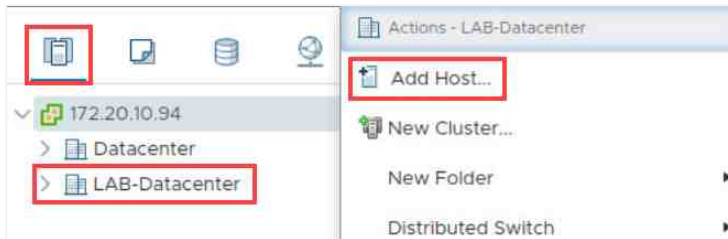


10. Leave the VMware vSphere Client open for the next task.

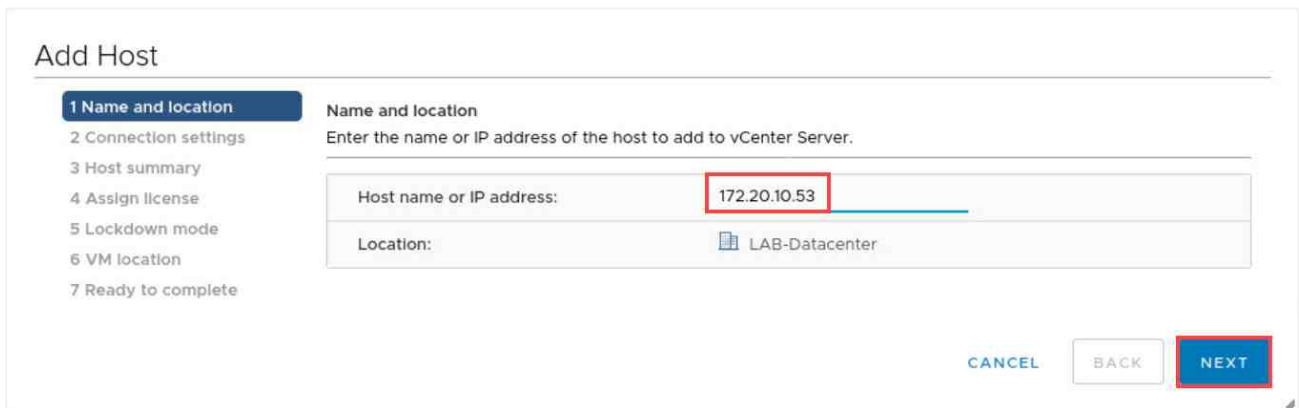
2 Add your ESXi host to the vCenter Server Inventory

In this task, you will add an ESXi host to the vCenter Server inventory.

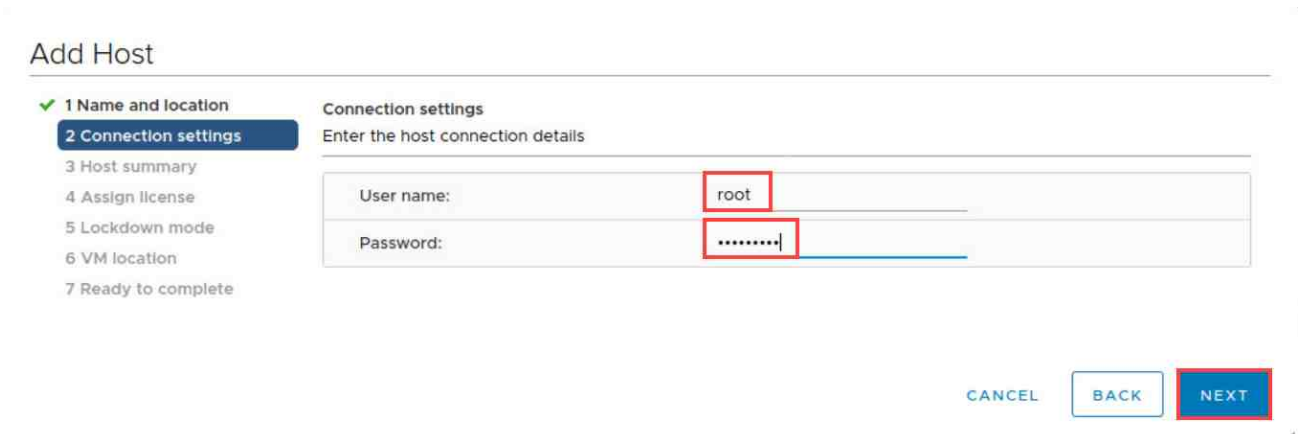
1. In the navigator pane, verify that Hosts and Clusters is selected. Right-click LAB-Datacenter and select Add Host.



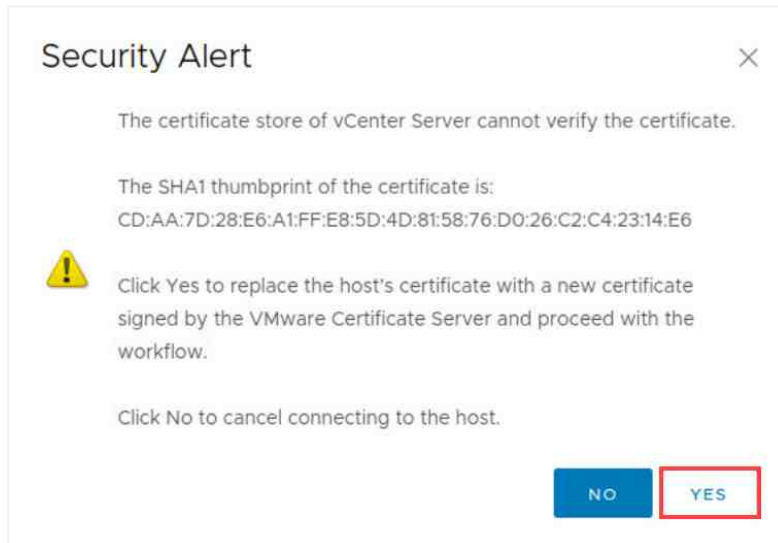
2. In the Add Host window, on the Name and location step, enter 172.20.10.53 and click Next.

A screenshot of the 'Add Host' wizard in vCenter Server. The 'Name and location' step is active. The 'Host name or IP address' field contains '172.20.10.53' and the 'Location' dropdown is set to 'LAB-Datacenter'. The 'NEXT' button is highlighted with a red box. The wizard steps on the left are: 1 Name and location, 2 Connection settings, 3 Host summary, 4 Assign license, 5 Lockdown mode, 6 VM location, 7 Ready to complete.

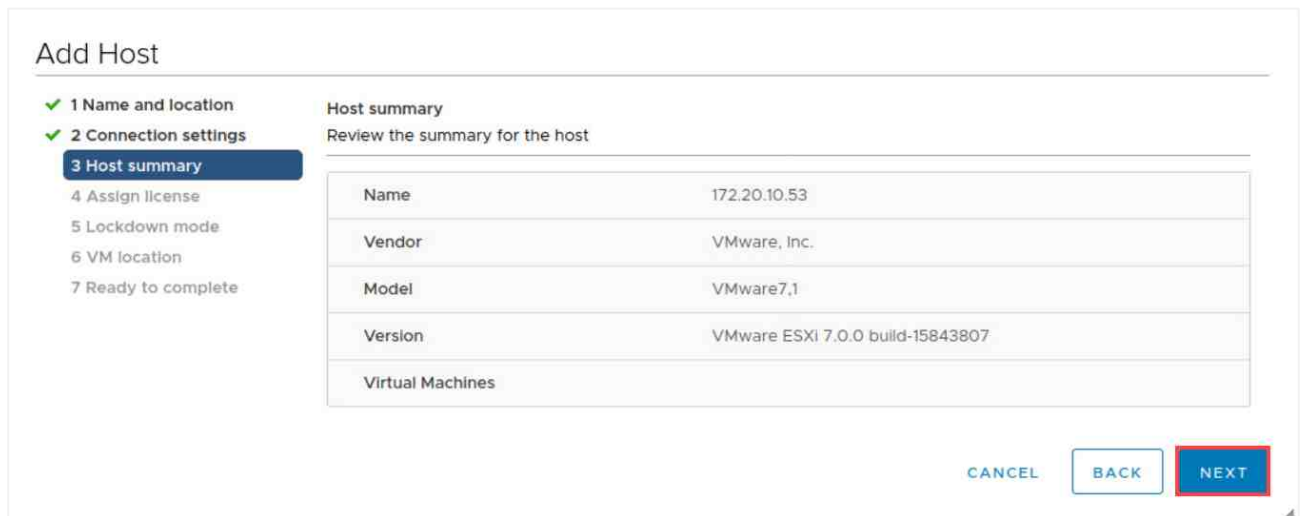
3. On the Connection settings step, enter root as the Username and Training\$ as the Password. Click Next.

A screenshot of the 'Add Host' wizard in vCenter Server, showing the 'Connection settings' step. The 'User name' field contains 'root' and the 'Password' field contains 'Training\$'. The 'NEXT' button is highlighted with a red box. The wizard steps on the left are: 1 Name and location, 2 Connection settings, 3 Host summary, 4 Assign license, 5 Lockdown mode, 6 VM location, 7 Ready to complete.

- When prompted with a security alert stating that the certificate store of vCenter Server cannot verify the certificate, click YES to proceed.



- Once validated, on the Host summary step, review the information and click Next.



6. On the Assign license step, select the radio button for VMware ESXi and click Next.

Add Host

- ✓ 1 Name and location
- ✓ 2 Connection settings
- ✓ 3 Host summary
- 4 Assign license**
- 5 Lockdown mode
- 6 VM location
- 7 Ready to complete

Assign license
Assign an existing or a new license to this host

	License	License Key	Product	Usage	Capacity
☐	Evaluation License	--	--	--	--
☒	VMware ESXi	0MOC2-NT203-...	VMware vSphere 7 Ent...	• 6 CPUs ...	8 CPUs (u...

Assignment Validation for VMware ESXi

⚠ Some features will become unavailable. [Details...](#)

[CANCEL](#) [BACK](#) [NEXT](#)

7. On the Lockdown mode step, accept the default, Disabled, and click Next.

Add Host

- ✓ 1 Name and location
- ✓ 2 Connection settings
- ✓ 3 Host summary
- ✓ 4 Assign license
- 5 Lockdown mode**
- 6 VM location
- 7 Ready to complete

Lockdown mode
Specify whether to enable lockdown mode on the host

When enabled, lockdown mode prevents remote users from logging directly into this host. The host will only be accessible through local console or an authorized centralized management application.

If you are unsure what to do, leave lockdown mode disabled. You can configure lockdown mode later by editing Security Profile in host settings.

☒ **Disabled**

☐ **Normal**
The host is accessible only through the local console or vCenter Server.

☐ **Strict**
The host is accessible only through vCenter Server. The Direct Console UI service is stopped.

[CANCEL](#) [BACK](#) [NEXT](#)

8. On the VM location step, verify LAB-Datacenter is showing and click Next.

Add Host

- ✓ 1 Name and location
- ✓ 2 Connection settings
- ✓ 3 Host summary
- ✓ 4 Assign license
- ✓ 5 Lockdown mode
- 6 VM location**
- 7 Ready to complete

VM location

LAB-Datacenter

CANCEL BACK NEXT

9. On the Ready to complete step, review the information and click Finish.

Add Host

- ✓ 1 Name and location
- ✓ 2 Connection settings
- ✓ 3 Host summary
- ✓ 4 Assign license
- ✓ 5 Lockdown mode
- ✓ 6 VM location
- 7 Ready to complete**

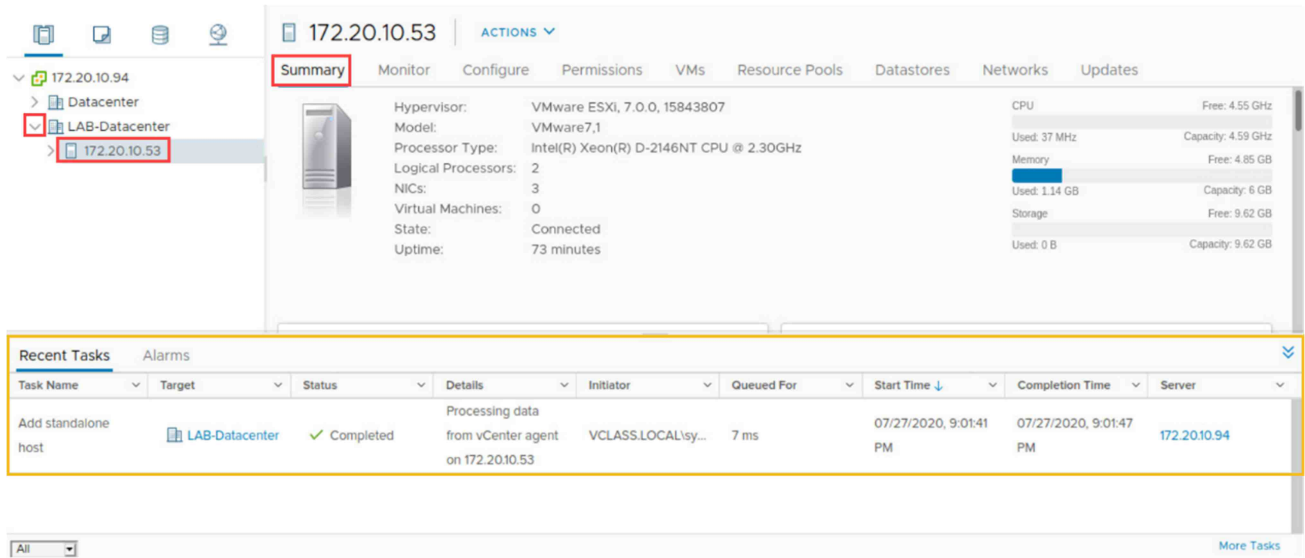
Ready to complete

Click Finish to add the host

Name	172.20.10.53
Location	LAB-Datacenter
Version	VMware ESXi 7.0.0 build-15843807
License	VMware ESXi
Networks	VM Network
Datastores	NFS-1
Lockdown mode	Disabled
VM location	LAB-Datacenter

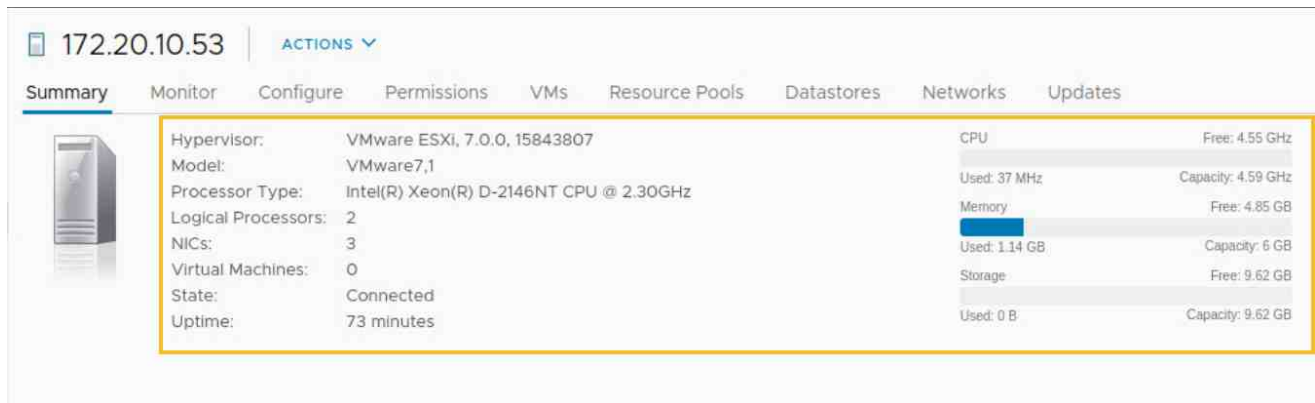
CANCEL BACK FINISH

10. Notice the progression in the Recent Tasks pane. Once completed, the ESXi host will appear underneath the LAB-Datacenter object in the navigator pane. Expand the LAB-Datacenter object and select 172.20.10.53, followed by clicking on the Summary tab in the main workspace.



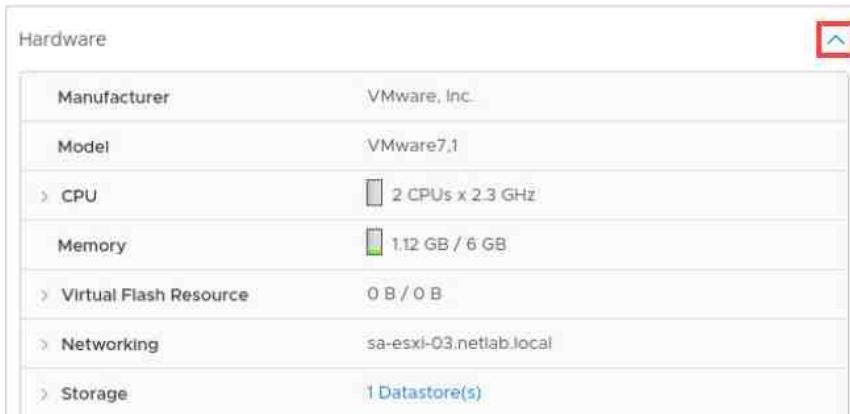
Task Name	Target	Status	Details	Initiator	Queued For	Start Time	Completion Time	Server
Add standalone host	LAB-Datacenter	Completed	Processing data from vCenter agent on 172.20.10.53	VCLASS.LOCAL\sy...	7 ms	07/27/2020, 9:01:41 PM	07/27/2020, 9:01:47 PM	172.20.10.94

11. Notice the ESXi host information such as CPU, memory, storage, NIC, and virtual machines.



Hypervisor:	VMware ESXi, 7.0.0, 15843807	CPU	Free: 4.55 GHz
Model:	VMware7,1	Used: 37 MHz	Capacity: 4.59 GHz
Processor Type:	Intel(R) Xeon(R) D-2146NT CPU @ 2.30GHz	Memory	Free: 4.85 GB
Logical Processors:	2	Used: 1.14 GB	Capacity: 6 GB
NICs:	3	Storage	Free: 9.62 GB
Virtual Machines:	0	Used: 0 B	Capacity: 9.62 GB
State:	Connected		
Uptime:	73 minutes		

12. Click the arrow next to the Hardware pane to view the hardware details of the ESXi host.

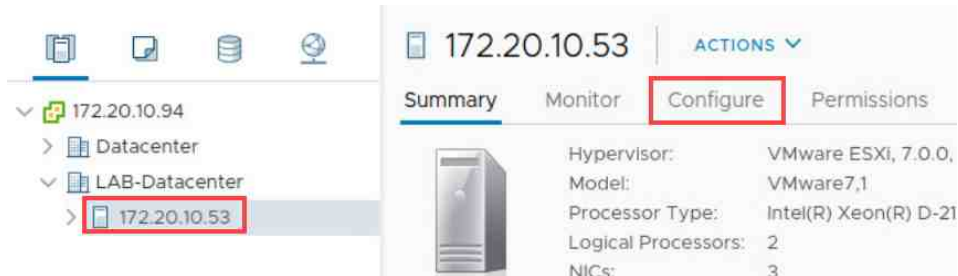


13. Leave the VMware vSphere Client open and continue to the next task.

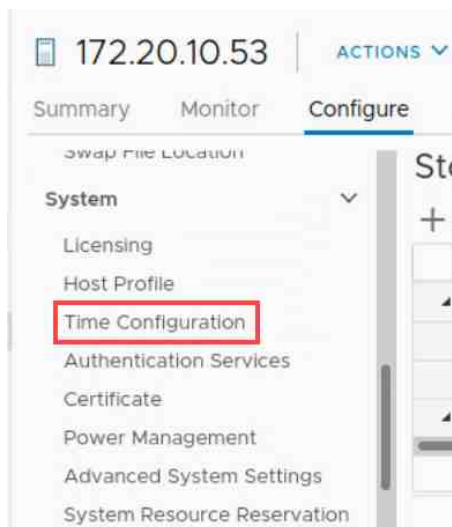
3 Configure ESXi host as an NTP Client

In this task, you will configure the ESXi host to use Network Time Protocol (NTP) to maintain accurate time and date.

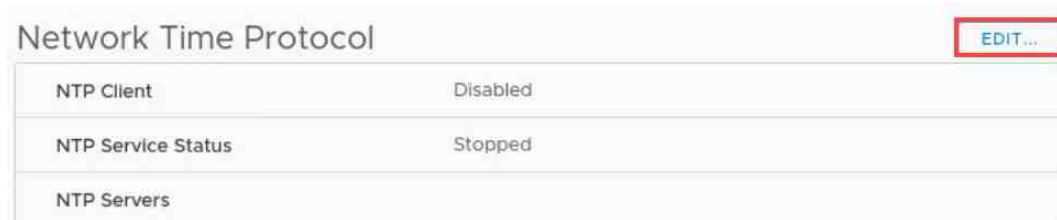
1. Select 172.20.10.53 (sa-esxi-03) in the navigator pane and click on the Configure tab in the main workspace.



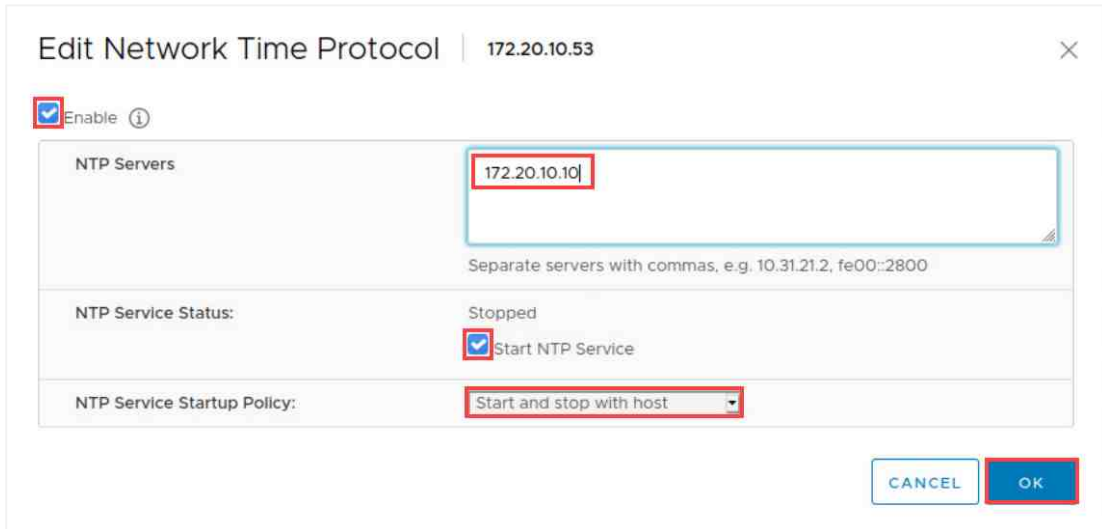
2. Under System in the middle pane (scroll down), select Time Configuration and view the current settings.



3. Scroll down to Network Time Protocol and click Edit.



4. In the Network Time Protocol window, select Enable to use the Network Time Protocol (NTP). In the NTP Servers text box, enter 172.20.10.10, followed by checking the checkbox for Start NTP service. In the NTP Service Startup Policy, select Start and Stop with host. Click OK.



172.20.10.53

☒ Enable ⓘ

NTP Servers: 172.20.10.10

Separate servers with commas, e.g. 10.31.21.2, fe00::2800

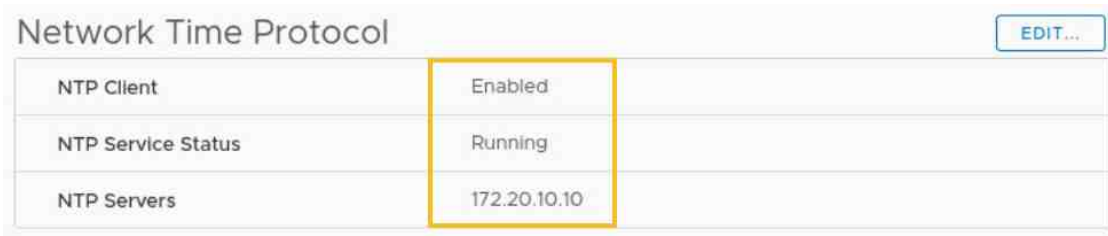
NTP Service Status: Stopped

☒ Start NTP Service

NTP Service Startup Policy: Start and stop with host

CANCEL OK

5. In the Network Time Protocol pane, verify that the NTP client appears as Enabled and the NTP service is Running from the NTP Server of 172.20.10.10.



Network Time Protocol EDIT...

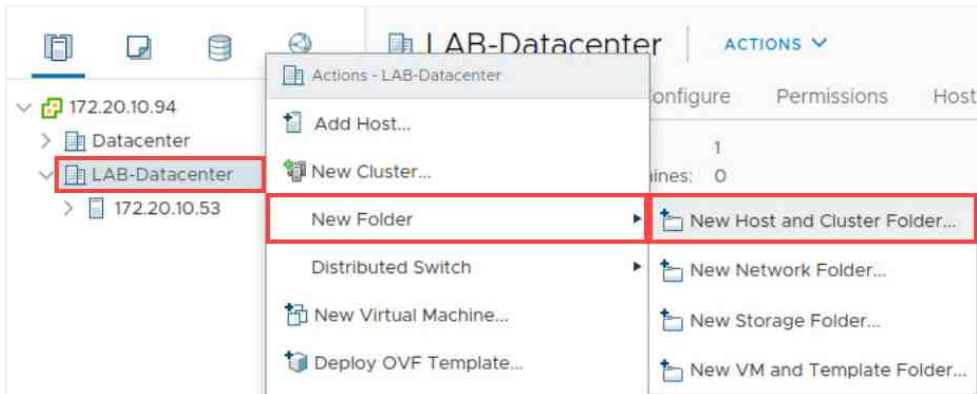
NTP Client	Enabled
NTP Service Status	Running
NTP Servers	172.20.10.10

6. Leave the VMware vSphere Client open and continue to the next task.

4 Create a folder for an ESXi host

In this task, you will create a folder called LAB-Servers and add 172.20.10.53 (sa-esxi-03.vclass.local) to it.

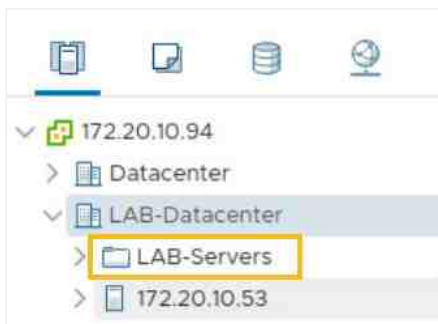
1. From the Hosts and Clusters tab, right-click LAB-Datacenter and select New Folder > New Host and Cluster Folder.



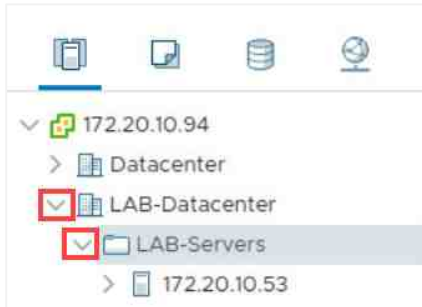
2. In the New Folder window, type LAB-Servers for the name and click OK.



3. Verify that LAB-Servers appear in the navigation pane.



4. In the navigator pane, drag 172.20.10.53 (sa-esxi-03) to the LAB-Servers folder. Once finished, expand the LAB-Servers folder to confirm that sa-esxi-03.vclass.local host now appears inside the folder.

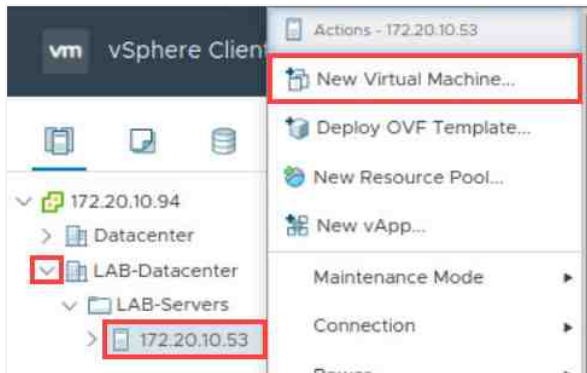


5. Leave the VMware vSphere Client open and continue to the next task.

5 Create a folder for a Virtual Machine and Virtual Machine Templates

In this task, you will create virtual machine and template folders.

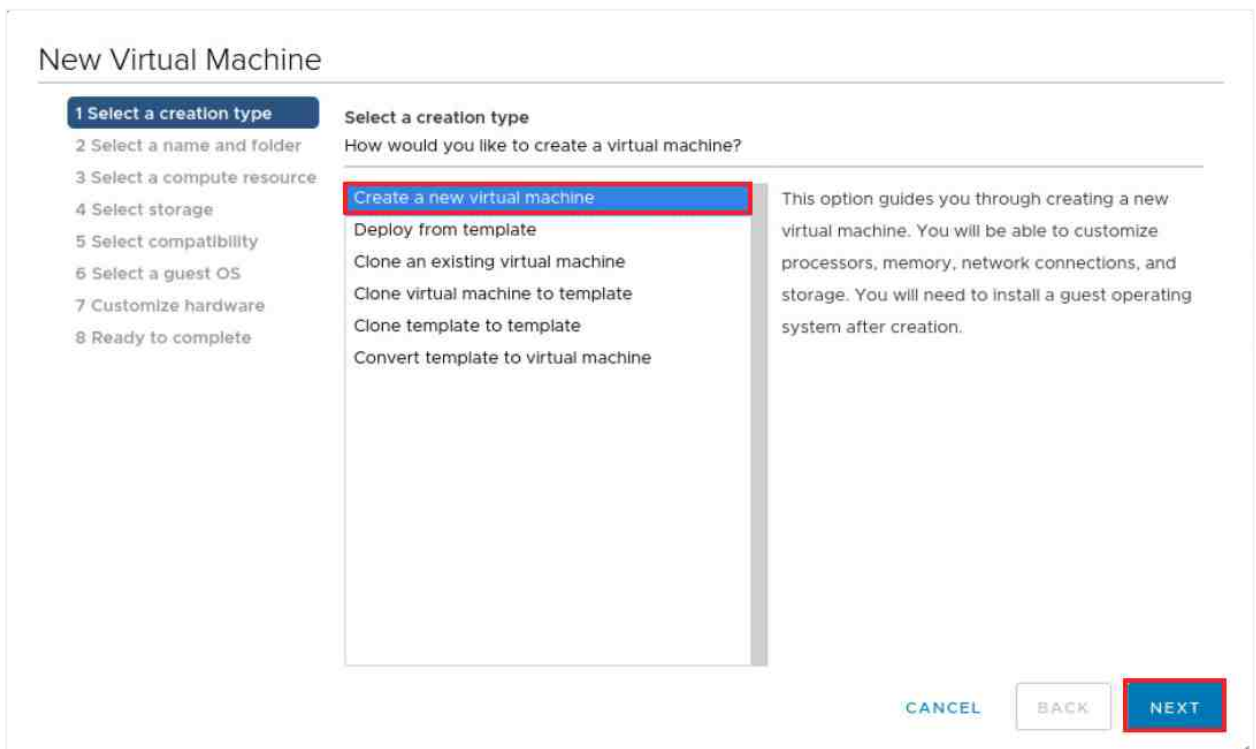
1. Expand the LAB-Datacenter object in the navigator pane. Right-click on 172.20.10.53 and select New Virtual Machine.



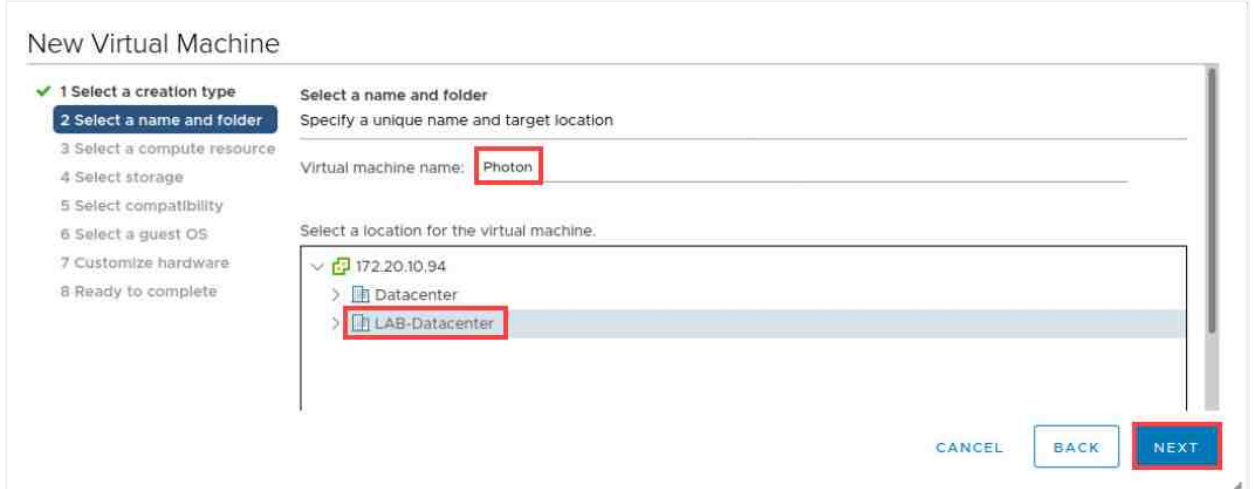
Please Note

To complete this task, you will create a new virtual machine on sa-esxi-03 (172.20.10.53) due to sa-esxi-03 not currently having any virtual machines residing on the host.

2. In the New Virtual Machine window, on the Select a creation type step, select Create a new virtual machine and click Next.

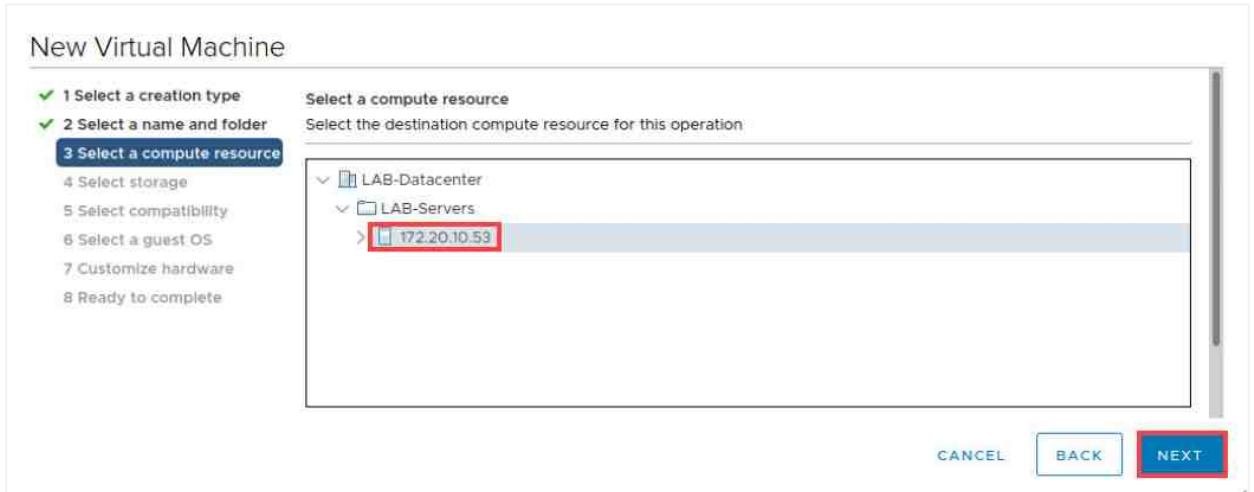


- On the Select a name and folder step, enter `Photon` as the name and ensure that LAB-Datacenter is selected for the location. Click Next.



The screenshot shows the 'New Virtual Machine' wizard at step 2, 'Select a name and folder'. The left sidebar lists steps 1 through 8, with step 2 highlighted. The main area has two sections: 'Select a name and folder' and 'Select a location for the virtual machine'. In the first section, the 'Virtual machine name' field contains 'Photon'. In the second section, a tree view shows the hierarchy: '172.20.10.94' > 'Datacenter' > 'LAB-Datacenter', with 'LAB-Datacenter' selected. At the bottom right are 'CANCEL', 'BACK', and 'NEXT' buttons, with 'NEXT' highlighted.

- On the Select a compute resource step, ensure that 172.20.10.53 is selected and click Next.



The screenshot shows the 'New Virtual Machine' wizard at step 3, 'Select a compute resource'. The left sidebar lists steps 1 through 8, with step 3 highlighted. The main area has two sections: 'Select a compute resource' and 'Select the destination compute resource for this operation'. In the second section, a tree view shows the hierarchy: 'LAB-Datacenter' > 'LAB-Servers' > '172.20.10.53', with '172.20.10.53' selected. At the bottom right are 'CANCEL', 'BACK', and 'NEXT' buttons, with 'NEXT' highlighted.

5. On the Select storage step, select NFS-1 to be the local datastore and click Next.

New Virtual Machine

- ✓ 1 Select a creation type
- ✓ 2 Select a name and folder
- ✓ 3 Select a compute resource
- 4 Select storage**
- 5 Select compatibility
- 6 Select a guest OS
- 7 Customize hardware
- 8 Ready to complete

Select storage
Select the storage for the configuration and disk files

☐ Encrypt this virtual machine (Requires Key Management Server)

VM Storage Policy: Datastore Default

Name	Capacity	Provisioned	Free	Type	Cluster
NFS-1	9.62 GB	0 B	9.62 GB	NFS v3	

Compatibility
✓ Compatibility checks succeeded.

CANCEL
BACK
NEXT

6. On the Select compatibility step, select ESXi 7.0 and later. Click Next.

New Virtual Machine

- ✓ 1 Select a creation type
- ✓ 2 Select a name and folder
- ✓ 3 Select a compute resource
- ✓ 4 Select storage
- 5 Select compatibility**
- 6 Select a guest OS
- 7 Customize hardware
- 8 Ready to complete

Select compatibility
Select compatibility for this virtual machine depending on the hosts in your environment

The host or cluster supports more than one VMware virtual machine version. Select a compatibility for the virtual machine.

Compatible with: ESXi 7.0 and later ⓘ

This virtual machine uses hardware version 17, which provides the best performance and latest features available in ESXi 7.0

CANCEL
BACK
NEXT

7. On the Select a guest OS step, select Linux as the Guest OS Family and VMware Photon OS (64-bit) as the Guest OS Version. Click Next.

New Virtual Machine

- ✓ 1 Select a creation type
- ✓ 2 Select a name and folder
- ✓ 3 Select a compute resource
- ✓ 4 Select storage
- ✓ 5 Select compatibility
- 6 Select a guest OS**
- 7 Customize hardware
- 8 Ready to complete

Select a guest OS
Choose the guest OS that will be installed on the virtual machine

Identifying the guest operating system here allows the wizard to provide the appropriate defaults for the operating system installation.

Guest OS Family: Linux

Guest OS Version: VMware Photon OS (64-bit)

Compatibility: ESXi 7.0 and later (VM version 17)

CANCEL
BACK
NEXT

8. On the Customize hardware, Virtual Hardware tab. Enter 5GB for the hard disk size and click Next.

New Virtual Machine

- ✓ 1 Select a creation type
- ✓ 2 Select a name and folder
- ✓ 3 Select a compute resource
- ✓ 4 Select storage
- ✓ 5 Select compatibility
- ✓ 6 Select a guest OS
- 7 Customize hardware**
- 8 Ready to complete

Customize hardware
Configure the virtual machine hardware

Virtual Hardware VM Options

[ADD NEW DEVICE](#)

> CPU	1	
> Memory	2	GB
> New Hard disk *	5	GB
> New SCSI controller *	VMware Paravirtual	
> New Network *	VM Network	☒ Connect...
> New CD/DVD Drive *	Client Device	☐ Connect...
> Video card *	Specify custom settings	
> Security Devices	Not Configured	
VMCI device		
New SATA Controller	New SATA Controller	
> Other	Additional Hardware	

Compatibility: ESXi 7.0 and later (VM version 17)

[CANCEL](#) [BACK](#) [NEXT](#)

9. On the Ready to complete step, review the information and click Finish.

New Virtual Machine

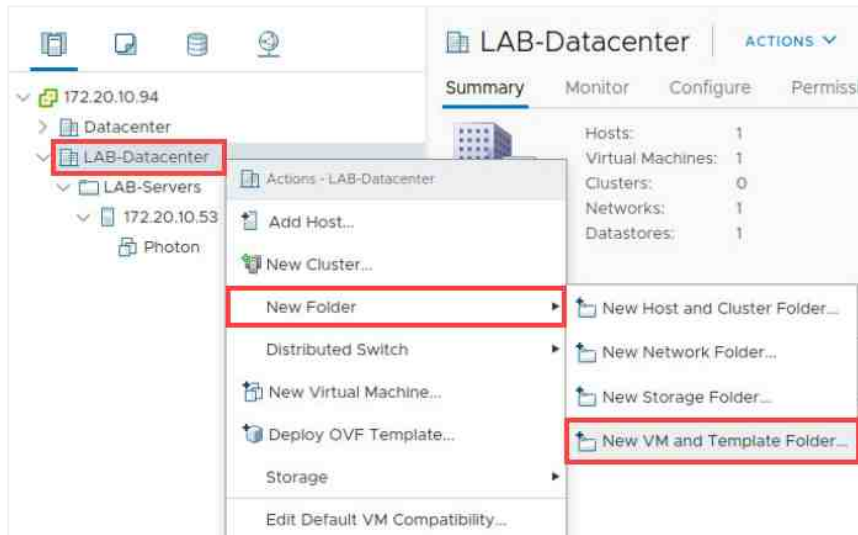
- ✓ 1 Select a creation type
- ✓ 2 Select a name and folder
- ✓ 3 Select a compute resource
- ✓ 4 Select storage
- ✓ 5 Select compatibility
- ✓ 6 Select a guest OS
- ✓ 7 Customize hardware
- 8 Ready to complete**

Ready to complete
Click Finish to start creation.

Virtual machine name	Photon
Folder	LAB-Datacenter
Host	172.20.10.53
Datastore	NFS-1
Guest OS name	VMware Photon OS (64-bit)
Virtualization Based Security	Disabled
CPUs	1
Memory	2 GB
NICs	1
NIC 1 network	VM Network
NIC 1 type	VMXNET 3
SCSI controller 1	VMware Paravirtual
Create hard disk 1	New virtual disk
Capacity	5 GB
Datastore	NFS-1

[CANCEL](#) [BACK](#) [FINISH](#)

10. From the Hosts and Clusters tab, right-click LAB-Datacenter and select New Folder > New VM and Template Folder.



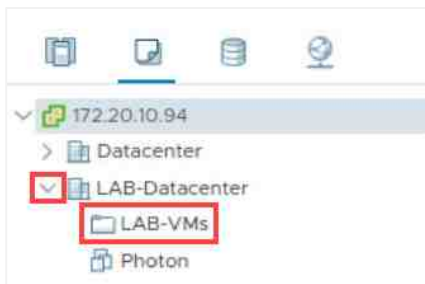
11. In the New Folder window, type LAB-VMs for the name and click OK.



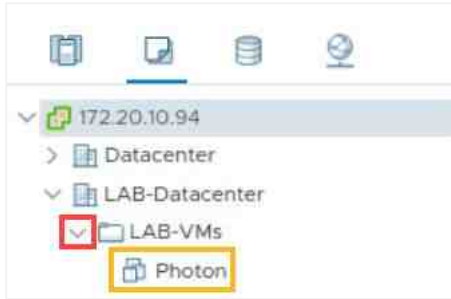
12. Navigate to the VMs and Templates tab in the navigator pane.



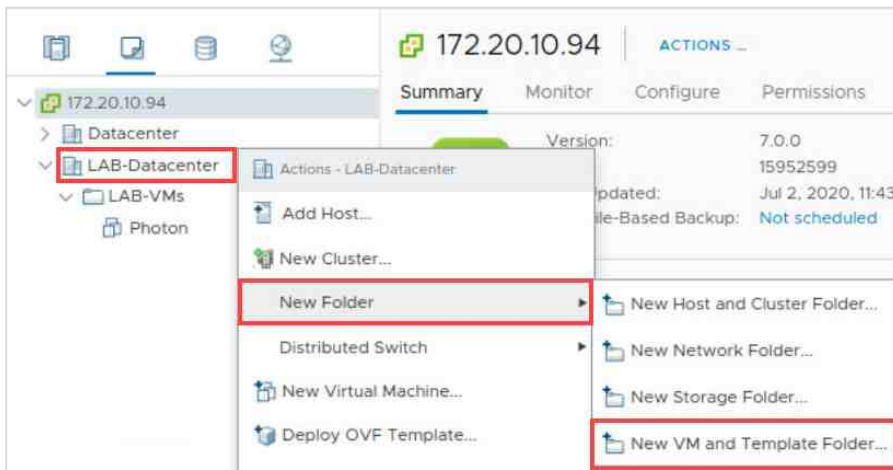
13. Expand the LAB-Datacenter object and notice the LAB-VMs folder appears in the navigator pane.



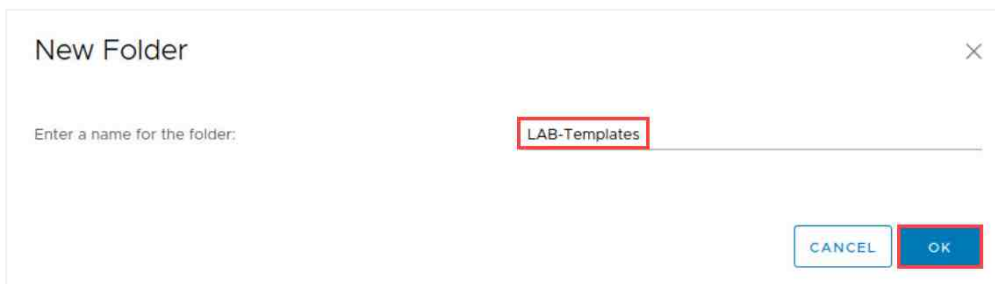
14. In the navigator pane, drag Photon to the LAB-VMs folder. Expand the LAB-VMs folder and verify Photon is showing.



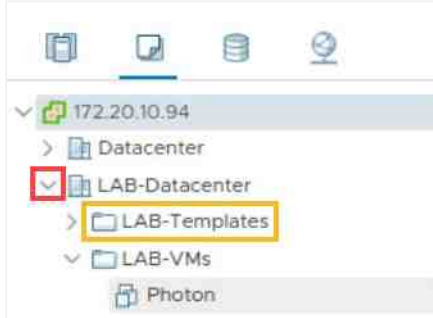
15. On the VMS and Templates tab, right-click LAB-Datacenter and select New Folder > New VM and Template Folder.



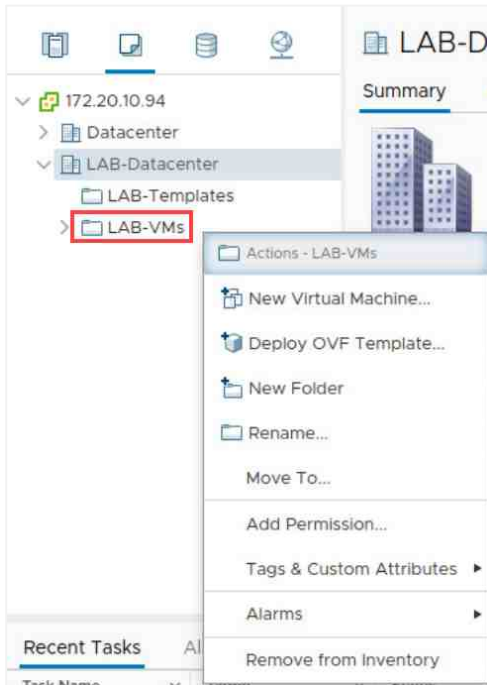
16. In the New Folder window, type LAB-Templates for the name and click OK.



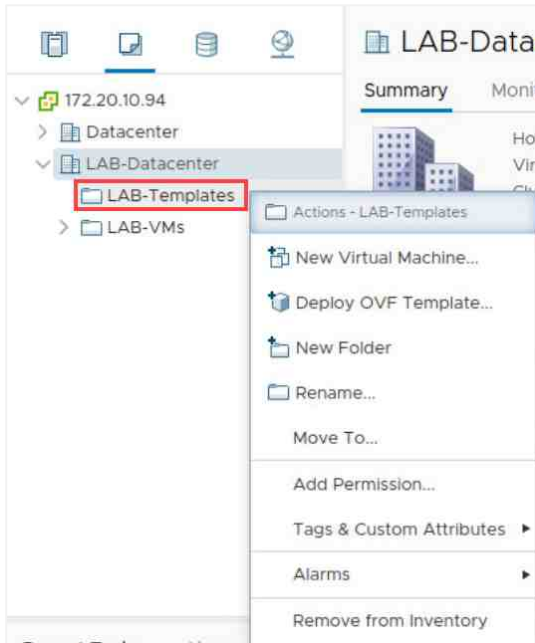
17. Expand the LAB-Datacenter object and notice the LAB-Templates folder appears in the navigator pane.



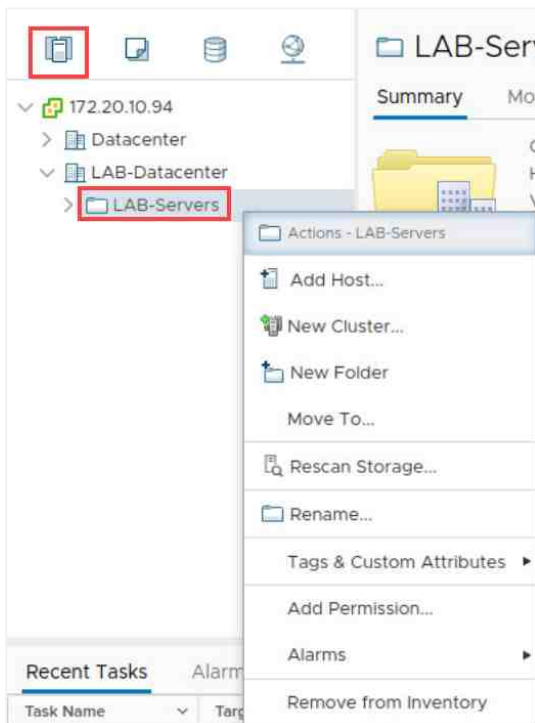
18. Right-click the LAB-VMs folder and notice the available menu commands in the dropdown menu.



19. Right-click the LAB-Templates folder and notice the available menu commands in the dropdown menu.



20. Navigate to the Hosts and Clusters tab in the navigator pane. Right-click the LAB-Servers folder and notice the available menu commands in the dropdown menu.



21. The lab is now complete; you may end the reservation.